Littlewood-type infinite-sum identity for modified Robbins polynomials (Theorem 1.1)

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Editorial note. Mathematical spans are unchanged native published TeX. Counters restore original theorem/definition/lemma and global equation numbers. Full 43-entry bibliography recovered from the official article HTML, preserving numbers and original keys; math uses official data-tex. No external figure or current cited proof dependency in the selected chain.

Editorial scope. One complete Theorem 1.1 with support; no mathematical repair, alternate-proof count or formal admission.

Allowed standard prerequisites (editorial).

- Pfaffian algebra. Source native 575–597, equation (6). Perfect-matching definition, skew-symmetric completion, simultaneous row/column operations and last-column expansion are retained.
- Polynomial interpolation and divisibility; formal geometric series. Complete Lemma 4.1 proof native 616–723 and main proof 727–757. Standard techniques applied to the explicit current identities, which are proved in full; no external current core or computation is invoked.

Source and scope. Ilse Fischer and Hans Höngesberg, *A Littlewood-type identity for Robbins polynomials*, Algebraic Combinatorics 9(3) (2026), 665–700, doi:10.5802/alco.495. Copyright the authors, 2026; CC BY 4.0. This extraction retains Theorem 1.1 as one counted result, Definition 2.1, the Pfaffian preliminaries, Lemma 4.1 with its full proof, and the complete first proof of Theorem 1.1. Original numbering and mathematical text are retained; the alternative proof sketch and subsequent consequences are not part of the counted package. The author source is unchanged. Only a standard article wrapper and bibliography recovered from the same official article have been supplied. This is author-material preparation, not independent mathematical certification. The geometric-series interpretation is the one used in the retained author proof.

Theorem 1.1. *Let n be a positive integer. Then*

$$\sum_{0\le k_1<\cdots<k_n} R^*_{(k_1,\dots,k_n)}(x_1,\dots,x_n;1,1,w) = \prod_{i=1}^{n}\frac{1}{1-x_i}\prod_{1\le i<j\le n}\frac{x_i+x_j+wx_ix_j}{x_j-x_i} \times \operatorname*{Pf}_{\chi_{\mathrm{even}}(n)\le i<j\le n}\left(\begin{cases}1, & i=0,\\ \frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}, & i\ge 1,\end{cases}\right) \quad (2)$$

where Pf *denotes the Pfaffian of an upper triangular array and* $\chi_{\mathrm{even}}(n)$ *equals* 1 *if* n *is even and* 0 *otherwise.*

2 Modified Robbins polynomials

To begin with, let us fix some notation. For any positive integer n, we denote by $\mathfrak{S}_n$ the symmetric group of order n. We define the *antisymmetrizer* $\mathbf{ASym}_{x_1,\dots,x_n}$ to be

$$\mathbf{ASym}_{x_1,\dots,x_n} F(x_1,\dots,x_n) \coloneqq \sum_{\sigma\in\mathfrak{S}_n}\operatorname{sgn}(\sigma)F(x_{\sigma(1)},\dots,x_{\sigma(n)}).$$

The antisymmetrizer is needed to define the modified Robbins polynomials, which are one of the central objects of this paper.

Definition 2.1. Let $\boldsymbol{k} = (k_1, \dots, k_n)$ be a sequence of integers. We define the *modified Robbins polynomials* $R^*_{\boldsymbol{k}}(x_1, \dots, x_n; u, v, w)$ to be

$$\frac{\mathbf{ASym}_{x_1,\dots,x_n}\left[\prod_{1\le i<j\le n}(ux_ix_j + v + wx_i)\prod_{i=1}^{n} x_i^{k_i}\right]}{\prod_{1\le i<j\le n}(x_j - x_i)}.$$

4 Proof of Theorem 1.1

The Littlewood-type identity for modified Robbins polynomials involves the notion of Pfaffians which we recall in the following before proving Theorem 1.1:

For every triangular array $A = (a_{i,j})_{1\le i<j\le 2n}$, the Pfaffian $\mathrm{Pf}(A)$ is defined as

$$\sum_{\{(i_1,j_1),\dots,(i_n,j_n)\}} \mathrm{sgn}(i_1 j_1 \dots i_n j_n) a_{i_1,j_1} \cdots a_{i_n,j_n},$$

where we sum over all perfect matchings $\{(i_1, j_1), \dots, (i_n, j_n)\}$ of $\{1, \dots, 2n\}$ such that $i_1 < \cdots < i_n$ and $i_k < j_k$ for all $1 \le k \le n$.

Every triangular array $A = (a_{i,j})_{1\le i<j\le 2n}$ is the upper triangular part of a uniquely determined $2n \times 2n$ skew-symmetric matrix M, which we call the *skew-symmetric completion* of A. A matrix $M = (m_{i,j})_{1\le i,j\le 2n}$ is skew-symmetric if $m_{i,j} = -m_{j,i}$ for all $1 \le i, j \le 2n$; in particular, all diagonal entries are 0. It is well known that $\mathrm{Pf}(A)^2 = \det(M)$. On that account, we will repeatedly change perspectives when dealing with Pfaffians as we will consider either the triangular array or the skew-symmetric matrix that underlies the Pfaffian.

We recall the main methods for manipulating Pfaffians used in the present paper:

- Regarding elementary row or column operations on the underlying matrix, we perform corresponding row and column operations at the same time. Thus, the resulting matrix remains skew-symmetric. In particular, simultaneously swapping corresponding rows and columns multiplies the Pfaffian by -1.
- Expanding the Pfaffian along the last column yields

$$\mathop{\mathrm{Pf}}_{1\le i<j\le 2n}(a_{i,j}) = \sum_{k=1}^{2n-1}(-1)^{k+1} a_{k,2n} \mathop{\mathrm{Pf}}_{\substack{1\le i<j\le 2n\\ i,j\ne k}}(a_{i,j}).$$

- For all $c_1, \dots, c_{2n}$, it holds that

$$\mathop{\mathrm{Pf}}_{1\le i<j\le 2n}(c_i c_j a_{i,j}) = \prod_{i=1}^{2n} c_i \mathop{\mathrm{Pf}}_{1\le i<j\le 2n}(a_{i,j}). \tag{6}$$

  Thus, for a given i, setting $c_i = c$ and all other $c_1, \dots, c_{i-1}, c_{i+1}, \dots, c_{2n}$ equal to 1 corresponds to multiplying the ith row and the ith column of the underlying matrix by c.

In preparation for the proof of Theorem 1.1, we show the following lemma which will be used for certain row and column operations:

Lemma 4.1. *The following identities hold:*

(i) For each positive integer n, it holds

$$\sum_{j=1}^{n}\left(\frac{1+x_j}{x_j}\prod_{\substack{1\le p\le n\\ p\ne j}}\frac{1-x_jx_p}{x_j-x_p}\right) + (-1)^n\prod_{k=1}^{n}\frac{1}{x_k} = 1.$$

(ii) For each positive integer n and i satisfying $1 \le i \le n$, it holds

$$(-1)^n\sum_{j=1}^{n}\left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\cdot\frac{1+x_j}{x_j}\prod_{\substack{1\le p\le n\\ p\ne j}}\frac{1-x_jx_p}{x_j-x_p}\right)$$
$$+\frac{1-x_i}{x_i}\prod_{\substack{1\le p\le n\\ p\ne i}}\frac{1+wx_i+x_ix_p}{x_i+x_p+wx_ix_p} = (-1)^n + \prod_{k=1}^{n}\frac{1}{x_k}.$$

Proof. Let us first prove (i). After some straightforward manipulation, the expression in (i) becomes the following polynomial identity:

$$\sum_{j=1}^{n}\left((-1)^j(1+x_j)\prod_{\substack{1\le p\le n\\ p\neq j}}x_p(1-x_jx_p)\prod_{\substack{1\le p<q\le n\\ p,q\neq j}}(x_q-x_p)\right)$$
$$=\left(-1+(-1)^n\prod_{p=1}^{n}x_p\right)\prod_{1\le p<q\le n}(x_q-x_p). \quad (7)$$

Both sides are multivariate polynomials in $x_1,\dots,x_n$ with highest degree at most n in each variable. It is not difficult to see that both sides vanish if we set $x_s=x_t$ for any $s\neq t$. Furthermore, if $x_s=0$, both sides evaluate to

$$(-1)^s\prod_{\substack{1\le p\le n\\ p\neq s}}x_p\prod_{\substack{1\le p<q\le n\\ p,q\neq s}}(x_q-x_p),$$

resulting in n coinciding evaluations for each variable. Thus, both sides are equal up to a multiplicative factor independent of $x_1,\dots,x_n$. Since the coefficient of $x_n^n x_{n-1}^{n-1}\cdots x_1$, that is, the leading coefficient in the lexicographic order for $x_n>\cdots>x_1$, equals $(-1)^n$ on both sides, we have shown (7).

Regarding (ii), we transform the identity into

$$\sum_{j=1}^{n}\Bigg((-1)^j(1+(1+w)x_ix_j)(1+x_j)(x_j-x_i)$$
$$\times\prod_{\substack{1\le p\le n\\ p\neq j}}x_p\prod_{\substack{1\le p\le n\\ p\neq i,j}}(1-x_jx_p)(x_i+x_p+wx_ix_p)\prod_{\substack{1\le p<q\le n\\ p,q\neq j}}(x_q-x_p)\Bigg)$$
$$+(1-x_i)\prod_{\substack{1\le p\le n\\ p\neq i}}x_p(1+wx_i+x_ix_p)\prod_{1\le p<q\le n}(x_q-x_p)$$
$$=\left(1+(-1)^n\prod_{k=1}^{n}x_k\right)\prod_{\substack{1\le p\le n\\ p\neq i}}(x_i+x_p+wx_ix_p)\prod_{1\le p<q\le n}(x_q-x_p), \quad (8)$$

both sides of which are multivariate polynomials in $x_1,\dots,x_n$; the highest degree of x_s is at most $n+1$ if $s\neq i$ and $2n-1$ otherwise.

A straightforward computation shows that both sides vanish if we set $x_s=x_t$ for any $s\neq t$. Setting $x_s=0$ yields

$$(-1)^{s-1}x_i^2\prod_{\substack{1\le p\le n\\ p\neq i,s}}x_p(x_i+x_p+wx_ix_p)\prod_{\substack{1\le p<q\le n\\ p,q\neq s}}(x_q-x_p)$$

on both sides if $s\neq i$; otherwise, we obtain

$$(-1)^{i-1}\prod_{\substack{1\le p\le n\\ p\neq i}}x_p^2\prod_{\substack{1\le p<q\le n\\ p,q\neq i}}(x_q-x_p).$$

Next, we consider the evaluation for $x_s=-\frac{x_i}{1+wx_i}$ if $s\neq i$. The right-hand side of (8) clearly vanishes. The left-hand side, however, becomes

$$\frac{(-1)^s}{(1+wx_i)^{n+1}}x_i(x_i-1)(2+wx_i)(1-x_i+wx_i)(1+x_i+wx_i)$$
$$\times\prod_{\substack{1\le p\le n\\ p\neq s}}x_p\prod_{\substack{1\le p\le n\\ p\neq i,s}}(1+wx_i+x_ix_p)(x_i+x_p+wx_ix_p)\prod_{\substack{1\le p<q\le n\\ p,q\neq s}}(x_q-x_p)$$
$$-\frac{(-1)^s}{(1+wx_i)^{n+1}}x_i(x_i-1)(1-x_i+wx_i)(1+x_i+wx_i)$$
$$\times\prod_{\substack{1\le p\le n\\ p\neq s}}(x_i+x_p+wx_ix_p)\prod_{\substack{1\le p\le n\\ p\neq i,s}}x_p(1+wx_i+x_ix_p)\prod_{\substack{1\le p<q\le n\\ p,q\neq s}}(x_q-x_p).$$

After omitting common factors, we are left with $(2+wx_i)x_i-(x_i+x_i+wx_i^2)$, which equals zero.

Next, let $x_i=-\frac{x_s}{1+wx_s}$ for any $s\neq i$. It is again obvious that the right-hand side of (8) vanishes. We obtain for the left-hand side

$$\begin{aligned}\frac{(-1)^{s+i-[s<i]}}{(1+wx_s)^{2n-1}}x_s^2(1-x_s)(1+x_s)(2+wx_s)(1+x_s+wx_s)\\ \times\prod_{\substack{1\le p\le n\\ p\neq i,s}}x_p(1-x_sx_p)(x_p-x_s)(x_s+x_p+wx_sx_p)\prod_{\substack{1\le p<q\le n\\ p,q\neq i,s}}(x_q-x_p)\\ +\frac{(-1)^{i-1}}{(1+wx_s)^{2n-1}}(1+x_s+wx_s)\prod_{\substack{1\le p\le n\\ p\neq i}}x_p(1-x_sx_p)(x_s+x_p+wx_sx_p)\\ \times\prod_{\substack{1\le p<q\le n\\ p,q\neq i}}(x_q-x_p),\end{aligned}$$

where $[\cdot]$ denotes the *Iverson bracket*: For any proposition P, $[P]$ evaluates to 1 if P is true and to 0 if P is false. It is again not difficult to see that the previous expression is zero.

In total, we have now $n+1$ matching evaluations for x_s for any $s\neq i$ and $2n-1$ evaluations for x_i. As a consequence, we know that the difference between the left-hand side and the right-hand side of (8) is the product of

$$\prod_{p=1}^{n}x_p\prod_{1\le p<q\le n}(x_q-x_p)\prod_{\substack{1\le p\le n\\ p\neq i}}(x_i+x_p+wx_ix_p)^2 \tag{9}$$

and a rational expression in w. In fact, we want to show that this factor equals zero. To this end, we consider the specializations $(x_1,\dots,x_n)=(x_1,\frac{1}{x_1},x_3,\frac{1}{x_3},\dots,x_{n-1},\frac{1}{x_{n-1}})$ if n is even and $(x_1,\dots,x_n)=(x_1,\frac{1}{x_1},x_3,\frac{1}{x_3},\dots,x_{n-2},\frac{1}{x_{n-2}},-1)$ if n is odd for which (9) does not vanish. Given these specializations, we will show that the left-hand side of (8) coincides with the right-hand side, which completes the proof of (8).

First, we consider the left-hand side of (8). Due to the factor $\prod_{\substack{1\le p\le n\\ p\neq i,j}}(1-x_jx_p)$, we see that the first sum vanishes unless $j=i+1$ if i is odd and $j=i-1$ if i is even. Let us assume the latter. Thus, it suffices to only keep the $(i-1)$th summand in the first sum. After some manipulation, the left-hand side of (8) reduces to

$$\begin{aligned}x_i(1+(1+w)x_{i-1}x_i)(1+x_{i-1})(x_i-x_{i-1})\\ \times\prod_{\substack{1\le p\le n\\ p\neq i-1,i}}x_p(1-x_{i-1}x_p)(x_i+x_p+wx_ix_p)(x_p-x_i)\prod_{\substack{1\le p<q\le n\\ p,q\neq i-1,i}}(x_q-x_p)\\ +x_{i-1}(1-x_i)(1+wx_i+x_{i-1}x_i)(x_i-x_{i-1})\\ \times\prod_{\substack{1\le p\le n\\ p\neq i-1,i}}x_p(1+wx_i+x_ix_p)(x_p-x_i)(x_p-x_{i-1})\prod_{\substack{1\le p<q\le n\\ p,q\neq i-1,i}}(x_q-x_p).\end{aligned}$$

The right-hand side of (8), however, can be recasted as

$$\begin{aligned}(x_{i-1}+x_i+wx_{i-1}x_i)(x_i-x_{i-1})\\ \times\left(1+(-1)^n\prod_{k=1}^{n}x_k\right)\prod_{\substack{1\le p\le n\\ p\neq i-1,i}}(x_i+x_p+wx_ix_p)(x_p-x_i)(x_p-x_{i-1})\prod_{\substack{1\le p<q\le n\\ p,q\neq i-1,i}}(x_q-x_p).\end{aligned}$$

After cancelling out common factors in the two previous expressions, it remains to show that

$$\begin{aligned}x_i(1+(1+w)x_{i-1}x_i)(1+x_{i-1})\prod_{\substack{1\le p\le n\\ p\neq i-1,i}}x_p(1-x_{i-1}x_p)(x_i+x_p+wx_ix_p)\\ +x_{i-1}(1-x_i)(1+wx_i+x_{i-1}x_i)\prod_{\substack{1\le p\le n\\ p\neq i-1,i}}x_p(1+wx_i+x_ix_p)(x_p-x_{i-1})\end{aligned}$$

equals

$$(x_{i-1}+x_i+wx_{i-1}x_i)\left(1+(-1)^n\prod_{k=1}^{n}x_k\right)\prod_{\substack{1\le p\le n\\ p\neq i-1,i}}(x_i+x_p+wx_ix_p)(x_p-x_{i-1})$$

for the specializations under consideration. For this purpose, let $k\neq i$ be even and consider the products

- $x_{k-1}(1-x_{i-1}x_{k-1})(x_i+x_{k-1}+wx_ix_{k-1})\cdot x_k(1-x_{i-1}x_k)(x_i+x_k+wx_ix_k)$,
- $x_{k-1}(1+wx_i+x_ix_{k-1})(x_{k-1}-x_{i-1})\cdot x_k(1+wx_i+x_ix_k)(x_k-x_{i-1})$,
- $(x_i+x_{k-1}+wx_ix_{k-1})(x_{k-1}-x_{i-1})\cdot(x_i+x_k+wx_ix_k)(x_k-x_{i-1})$,

all of which evaluate to

$$\frac{1}{x_{k-1}^2x_i^2}(x_{k-1}-x_{i-1})(1-x_{i-1}x_{k-1})(w+x_{i-1}+x_{k-1})(1+wx_{k-1}+x_{i-1}x_{k-1})$$

when setting $x_i\mapsto\frac{1}{x_{i-1}}$ and $x_k\mapsto\frac{1}{x_{k-1}}$. In addition, in the case of n being odd, one can easily check that all of the following expressions

- $x_n(1-x_{i-1}x_n)(x_i+x_n+wx_ix_n)$,
- $x_n(1+wx_i+x_ix_n)(x_n-x_{i-1})$,
- $(x_i+x_n+wx_ix_n)(x_n-x_{i-1})$

are equal to $(1+x_{i-1})(1-x_i+wx_i)$ if setting $x_n=-1$.

Moreover, we see that

$$x_i(1+(1+w)x_{i-1}x_i)(1+x_{i-1})+x_{i-1}(1-x_i)(1+wx_i+x_{i-1}x_i)$$

and

$$(x_{i-1}+x_i+wx_{i-1}x_i)\left(1+(-1)^n\prod_{k=1}^n x_k\right)$$

both specialize to

$$\frac{2}{x_{i-1}}(1+wx_{i-1}+x_{i-1}^2).$$

This finally concludes the proof of (8) for even i. An analogous argument holds in the case of i being odd. □

We are now ready to proceed with the main proof of Theorem 1.1.

First proof of Theorem 1.1. We prove the statement by induction on n, showing that both sides of (2) satisfy the same recurrence relation.

To start with, recall that $R^*_{\boldsymbol{k}}(x_1,\dots,x_n;1,1,w)$ is given by

$$\frac{\mathbf{ASym}_{x_1,\dots,x_n}\left[\prod_{1\le i<j\le n}(x_ix_j+1+wx_i)\prod_{i=1}^n x_i^{k_i}\right]}{\prod_{1\le i<j\le n}(x_j-x_i)}.$$

Through geometric series evaluation, it follows that

$$F(x_1,\dots,x_n)\coloneqq\sum_{0\le k_1<\cdots<k_n}R^*_{(k_1,\dots,k_n)}(x_1,\dots,x_n;1,1,w)$$

equals

$$\frac{\mathbf{ASym}_{x_1,\dots,x_n}\left[\prod_{1\le i<j\le n}(x_ix_j+1+wx_i)\prod_{i=1}^n x_i^{i-1}(1-\prod_{j=i}^n x_j)^{-1}\right]}{\prod_{1\le i<j\le n}(x_j-x_i)} \tag{10}$$

since

$$\begin{aligned}\sum_{0\le k_1<\cdots<k_n}\prod_{i=1}^n x_i^{k_i}&=\prod_{i=1}^n x_i^{i-1}\sum_{0\le k_1\le\cdots\le k_n}\prod_{i=1}^n x_i^{k_i}\\&=\prod_{i=1}^n x_i^{i-1}\sum_{0\le k_1\le\cdots\le k_n}(x_1\dots x_n)^{k_1}(x_2\dots x_n)^{k_2-k_1}\cdots x_n^{k_n-k_{n-1}}\\&=\prod_{i=1}^n x_i^{i-1}\sum_{k_1=0}^{\infty}(x_1\dots x_n)^{k_1}\sum_{k_2=0}^{\infty}(x_2\dots x_n)^{k_2}\cdots\sum_{k_n=0}^{\infty}x_n^{k_n}=\prod_{i=1}^n\frac{x_i^{i-1}}{1-\prod_{j=i}^n x_j}.\end{aligned}$$

Thus, we obtain the recurrence relation

$$F(x_1,\dots,x_n)=\left(1-\prod_{i=1}^{n}x_i\right)^{-1}\sum_{k=1}^{n}\prod_{\substack{1\le i\le n\\ i\neq k}}\frac{(x_ix_k+1+wx_k)x_i}{x_i-x_k}F(x_1,\dots,\widehat{x_k},\dots,x_n) \tag{11}$$

with initial condition

$$F(x_1)=\frac{1}{1-x_1},$$

where $x_1,\dots,\widehat{x_k},\dots,x_n$ denotes the omission of x_k in $x_1,\dots,x_n$.

If $n=1$, then both sides of (2) clearly evaluate to $1/(1-x_1)$. Now consider the induction step from $n-1$ to n. We make a case distinction regarding the parity of n.

Let us first assume that n is even. Using the induction hypothesis and the recursion (11), it follows that the right-hand side of (2) equals

$$\left(1-\prod_{i=1}^{n}x_i\right)^{-1}\sum_{k=1}^{n}\prod_{\substack{1\le i\le n\\ i\neq k}}\frac{(x_ix_k+wx_k+1)x_i}{(x_i-x_k)(1-x_i)}$$
$$\times\prod_{\substack{1\le i<j\le n\\ i,j\neq k}}\frac{x_i+x_j+wx_ix_j}{x_j-x_i}\operatorname*{Pf}_{\substack{1\le i<j\le n+1\\ i,j\neq k}}\left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)\Bigg|_{x_{n+1}=1},$$

which we rewrite as

$$\left(1-\prod_{i=1}^{n}x_i\right)^{-1}\prod_{1\le i<j\le n}\frac{x_i+x_j+wx_ix_j}{x_j-x_i}\sum_{k=1}^{n}(-1)^{k+1}\left(\prod_{\substack{1\le i\le n\\ i\neq k}}\frac{1+wx_k+x_kx_i}{x_k+x_i+wx_kx_i}\right)$$
$$\times\prod_{\substack{1\le i\le n\\ i\neq k}}\frac{x_i}{1-x_i}\operatorname*{Pf}_{\substack{1\le i<j\le n+1\\ i,j\neq k}}\left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)\Bigg|_{x_{n+1}=1}.$$

Interpreting the sum as an expansion along the last column yields

$$\left(1-\prod_{i=1}^{n}x_i\right)^{-1}\prod_{1\le i<j\le n}\frac{x_i+x_j+wx_ix_j}{x_j-x_i}$$
$$\times\operatorname*{Pf}_{1\le i<j\le n+2}\left(\begin{array}{c|c|c}\left(\frac{x_i}{1-x_i}\frac{x_j}{1-x_j}\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & \left(\frac{x_i}{1-x_i}\right)_{1\le i\le n} & \left(\prod_{\substack{1\le p\le n\\ p\neq i}}\frac{1+wx_i+x_ix_p}{x_i+x_p+wx_ix_p}\right)_{1\le i\le n}\\ \hline & & 0\end{array}\right).$$

We extract $\prod_{i=1}^{n}\frac{x_i}{1-x_i}$ from the Pfaffian using (6) and split it up into two summands as follows:

$$\operatorname*{Pf}_{1\le i<j\le n+2}\left(\begin{array}{c|c|c}\left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (1)_{1\le i\le n} & \left(\frac{1-x_i}{x_i}\left(\prod_{\substack{1\le p\le n\\ p\neq i}}\frac{1+wx_i+x_ix_p}{x_i+x_p+wx_ix_p}\right)\right)_{1\le i\le n}\\ \hline & & -\prod_{k=1}^{n}\frac{1}{x_k}\end{array}\right)$$
$$+\prod_{i=1}^{n}\frac{1}{x_i}\operatorname*{Pf}_{1\le i<j\le n}\left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right).$$

This implies that the proof of the even case is complete if we show that

$$\operatorname*{Pf}_{1\le i<j\le n+2}\left(\begin{array}{c|c|c}\left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (-1)_{1\le i\le n} & \left(\frac{1-x_i}{x_i}\left(\prod_{\substack{1\le p\le n\\ p\neq i}}\frac{1+wx_i+x_ix_p}{x_i+x_p+wx_ix_p}\right)\right)_{1\le i\le n}\\ \hline & & \prod_{k=1}^{n}\frac{1}{x_k}\end{array}\right)$$
$$=\operatorname*{Pf}_{1\le i<j\le n}\left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right), \tag{12}$$

where we incorporated a negative sign into the Pfaffian on the left-hand side by multiplying the $(n+1)$th row and $(n+1)$th column by -1 using (6).

We perform the following elementary row and column operations on the matrix underlying the Pfaffian in the left-hand side of (12): For every $j \in \{1,\dots,n\}$, we add the jth column times

$$\frac{1+x_j}{x_j} \prod_{\substack{1\le p\le n\\ p\ne j}} \frac{1-x_jx_p}{x_j-x_p}$$

to the last column; we then proceed analogously for the rows. Due to Lemma 4.1, this results in the left-hand side of (12) being equal to

$$\operatorname*{Pf}_{1\le i<j\le n+2} \left(\begin{array}{c|c|c} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (-1)_{1\le i\le n} & \left(1+\prod_{k=1}^n \frac{1}{x_k}\right)_{1\le i\le n} \\ \hline & & 1 \end{array} \right).$$

We further manipulate the underlying matrix as follows: We multiply the $(n+1)$th column by $1+\prod_{i=1}^n \frac{1}{x_i}$ and add it to the $(n+2)$th column; then we proceed similarly with the last two rows. Thus, we obtain

$$\operatorname*{Pf}_{1\le i<j\le n+2} \left(\begin{array}{c|c|c} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (-1)_{1\le i\le n} & (0)_{1\le i\le n} \\ \hline & & 1 \end{array} \right).$$

Evaluating this Pfaffian along the last column, finally yields the right-hand side of (12).

On the other hand, let us assume that n is odd. Similar as before, applying the recursion (11) to the induction hypothesis and some manipulations imply the right-hand side of (2) being equal to

$$\left(1-\prod_{i=1}^n x_i\right)^{-1} \prod_{1\le i<j\le n} \frac{x_i+x_j+wx_ix_j}{x_j-x_i} \sum_{k=1}^n (-1)^{k+1} \left(\prod_{\substack{1\le i\le n\\ i\ne k}} \frac{1+wx_k+x_kx_i}{x_k+x_i+wx_kx_i} \right) \times \prod_{\substack{1\le i\le n\\ i\ne k}} \frac{x_i}{1-x_i} \operatorname*{Pf}_{\substack{1\le i<j\le n\\ i,j\ne k}} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)} \right).$$

We multiply the factor $\prod_{\substack{1\le i\le n\\ i\ne k}} \frac{x_i}{1-x_i}$ into the Pfaffian using (6) and then interpret the sum as an expansion along the last column, which yields

$$\left(1-\prod_{i=1}^n x_i\right)^{-1} \prod_{1\le i<j\le n} \frac{x_i+x_j+wx_ix_j}{x_j-x_i} \times \operatorname*{Pf}_{1\le i<j\le n+1} \left(\begin{array}{c|c} \left(\frac{x_i}{1-x_i}\frac{x_j}{1-x_j}\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & \left(\prod_{\substack{1\le p\le n\\ p\ne i}} \frac{1+wx_i+x_ix_p}{x_i+x_p+wx_ix_p}\right)_{1\le i\le n} \end{array} \right).$$

Next, we extract again the product $\prod_{i=1}^n \frac{x_i}{1-x_i}$ from the Pfaffian using (6) and recast it by enlarging the dimension of the underlying array as

$$\operatorname*{Pf}_{1\le i<j\le n+3} \left(\begin{array}{c|c|c|c} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (1)_{1\le i\le n} & \left(\frac{1-x_i}{x_i}\left(\prod_{\substack{1\le p\le n\\ p\ne i}} \frac{1+wx_i+x_ix_p}{x_i+x_p+wx_ix_p}\right)\right)_{1\le i\le n} & (0)_{1\le i\le n} \\ \hline & & 0 & -1 \\ \hline & & & 0 \end{array} \right);$$

we then split up the Pfaffian into the following sum:

$$\operatorname*{Pf}_{1\le i<j\le n+3} \left(\begin{array}{c|c|c|c} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (1)_{1\le i\le n} & \left(\frac{1-x_i}{x_i}\left(\prod_{\substack{1\le p\le n\\ p\ne i}} \frac{1+wx_i+x_ix_p}{x_i+x_p+wx_ix_p}\right)\right)_{1\le i\le n} & (0)_{1\le i\le n} \\ \hline & & 0 & -1 \\ \hline & & & -\prod_{k=1}^n \frac{1}{x_k} \end{array} \right) + \prod_{i=1}^n \frac{1}{x_i} \operatorname*{Pf}_{1\le i<j\le n+1} \left(\begin{array}{c|c} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (1)_{1\le i\le n} \end{array} \right).$$

After putting all together and some minor manipulations, it remains to show that

$$\operatorname*{Pf}_{1\le i<j\le n+3}\left(\begin{array}{c|c|c|c} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (1)_{1\le i\le n} & \left(\frac{1-x_i}{x_i}\left(\prod_{\substack{1\le p\le n\\ p\neq i}}\frac{1+wx_i+x_ix_p}{x_i+x_p+wx_ix_p}\right)\right)_{1\le i\le n} & (0)_{1\le i\le n} \\ \hline & 0 & 1 & \\ \hline & & \prod_{k=1}^n\frac{1}{x_k} & \end{array}\right)$$

$$=\operatorname*{Pf}_{1\le i<j\le n+1}\left(\begin{array}{c|c} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (1)_{1\le i\le n} \end{array}\right). \quad (13)$$

To this end, we perform similar elementary row and column operations as in the even case on the matrix underlying the Pfaffian in the left-hand side of the previous equation. We multiply the ith column by

$$-\frac{1+x_i}{x_i}\prod_{\substack{1\le p\le n\\ p\neq i}}\frac{1-x_ix_p}{x_i-x_p}$$

and add it to the penultimate column for every $i\in\{1,\dots,n\}$; proceed analogously for the rows. We then obtain for the left-hand side of (13) the following due to Lemma 4.1:

$$\operatorname*{Pf}_{1\le i<j\le n+3}\left(\begin{array}{c|c|c|c} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (1)_{1\le i\le n} & \left(-1+\prod_{k=1}^n\frac{1}{x_k}\right)_{1\le i\le n} & (0)_{1\le i\le n} \\ \hline & & 1+\prod_{k=1}^n\frac{1}{x_k} & 1 \\ \hline & & & \prod_{k=1}^n\frac{1}{x_k} \end{array}\right). \quad (14)$$

We continue with further row and column operations regarding (14): we add $1-\prod_{k=1}^n\frac{1}{x_k}$ times the $(n+1)$th column and $-(1+\prod_{k=1}^n\frac{1}{x_k})$ times the $(n+3)$th column to the $(n+2)$th column; similarly for the rows. Finally, we subtract the $(n+1)$ row and column from the $(n+2)$th row and column, respectively. We eventually obtain

$$\operatorname*{Pf}_{1\le i<j\le n+3}\left(\begin{array}{c|c|c|c} \left(\frac{(x_j-x_i)(1+(1+w)x_ix_j)}{(x_i+x_j+wx_ix_j)(1-x_ix_j)}\right)_{1\le i<j\le n} & (1)_{1\le i\le n} & (0)_{1\le i\le n+1} & (0)_{1\le i\le n+1} \\ \hline & & & 1 \end{array}\right).$$

Expanded along the last column, this expression finally coincides with the right-hand side of (13). □

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